

System Integration Guide

To: Flutter, Backend, and UI/UX Teams

1. Flutter Team Instructions

- **Setup:** We will export the Unity AI module as an Android Library. You need to implement the 'flutter_unity_widget' package to host the camera view within your app screens.
- **Receiving Data:** Implement the onUnityMessage(message) callback. When the patient presses the "Finish" button in the Unity view, our C# script will fire a serialized JSON string to this callback.
- **Payload Format:** {"exercise": "Squat", "reps": 12}
- **Action Required:** Parse this JSON string, append the current Patient_ID and Timestamp, and forward the complete payload to the Backend API.

2. Backend Team Instructions

- **Architecture:** You will not receive any direct HTTP requests from the Unity engine. All database communications will go exclusively through the Flutter application.
- **Database Schema Prep:** Ensure your API endpoint and database tables are ready to accept and store the following session summary parameters: Patient ID, Exercise Type, Repetition Count, and Session Timestamp.

3. UI/UX Team Instructions

- **Screen Layout:** The Unity widget will take over a portion (or the entirety) of the screen to show the device's front camera.
- **Overlay Elements:** The Unity view already renders its own real-time text overlays (Dynamic instructions like "Go Lower", "Good Hold", and a Rep Counter) plus a "Finish" button. Please design the surrounding Flutter UI to match this interactive camera view seamlessly.